MIL-C-45924(WC)

26 March 1974
SUPERSEDING
(See Section 6)

MILITARY SPECIFICATION

CONTROL PANEL, GUNNER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of gunner's control used to operate the M28A1 Helicopter Armament Subsystem. The control incorporates switches, panel lights and instruments to indicate weapons employed, operational (circuit) functions and percentage of remaining ammunition. The gunner's control also provides an override of pilot armament functions and for firing wing stores armament.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-13855 - Weapons, Small Arms and Aircraft Armament Subsystems, General Specification for.

MIL-P-14232 - Parts, Equipment and Tools for Army Materiel, Packaging and Packing of.

MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of.

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 - Quality Assurance Terms and Definitions.

FSC 1090

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DRAWINGS

U.S. Army Weapons Command
F11687874 - Control Box, Gunner.
IEL11687874 - Index of Inspection Equipment List.

(Product drawings referenced in this specification form a part of F11687874; Acceptance inspection equipment drawings form a part of IEL11687874.)

PUBLICATIONS

U.S. Army Weapons Command
P11687874 - Packaging Data Sheet for Control Panel, Gunners.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.2). Unless otherwise specified (see 6.2), the first article shall include a pilot pack (see 5.1).

3.2 Material, construction and design. Controls shall conform to the material, construction and design requirements specified herein, on Drawing F11687874 and in MIL-W-13855.

3.2.1 Weight. The weight of the control shall not exceed 3.3 pounds.

3.3 Functional requirements.

3.3.1 Power inputs. Controls shall be capable of meeting the requirements specified herein, when supplied with 24 to 30 volts direct current (vdc) and a minimum current of 3.0 amperes.

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3.3.2 Panel edge lights (DS7, DS8, DS9, DS12, DS13, and DS14). The six panel edge lights shall illuminate when 24-30 vdc is applied between pin Z (positive) and pin R of connector A5J1.

3.3.3 Switches.

3.3.3.1 Pilot override switch (S1). With the pilot override switch (S1) in the OFF position, there shall be an open circuit between pin $\overline{C}$ (positive) and the following pins: C, X, Y, V and L of connector A5J1. There shall be continuity between pins C, X and Y of connector A5J1 and also between pins V and L of connector A5J1. With the pilot override switch (S1) in the ON position, there shall be continuity between pin $\overline{C}$ and the following pins: C, X, Y, V and L of connector A5J1.

3.3.3.2 POINT/AREA fire switch (S3). With the fire switch (S3) in the POINT position, there shall be an open circuit between pins A and $\overline{E}$ of connector A5J1. With the fire switch in the AREA position there shall be continuity between pins A and $\overline{E}$ of connector A5J1.

3.3.3.3 Wing storage select switch (S4). With the wing storage select switch (S4) in the OFF position, there shall be an open circuit between pin S and pins T and U of connector A5J1. In the INBD position there shall be continuity between pins S and T of connector A5J1. In the OUTBD position there shall be continuity between pins S and U of connector A5J1.

3.3.3.4 Ammunition fire out switch (S5). With the fire out switch (S5) in the ON position, there shall be continuity between pin N and pin M of connector A5J1 and between pin $\overline{G}$ and pin $\overline{H}$ of connector A5J1. With the fire out switch in the OFF position, and open circuit shall exist between pin N and pin M and between pin $\overline{G}$ and pin $\overline{H}$ of connector A5J1.

3.3.3.5 Weapon switch (S6). With the weapon switch (S6) in the UNCLEAR position, there shall be continuity between pin G and pin W of connector A5J1. With the switch in the CLEAR position, an open circuit shall exist between pin G and pin W of connector A5J1.

3.3.4 Indicators.

3.3.4.1 Ammunition reserve indicators. The left and right rounds counter shall display the percentage of ammunition remaining in the respective magazine. Operation of a rounds counter from 100% to 0% shall require voltage pulses at the rate of 0 to 500 pulses per minute

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(minimum 15 milliseconds in length) at 24-30 vdc. Voltage pulses shall be applied to terminal D for the left counter and terminal P for the right counter with a common ground connection to terminal R. The diode connected across the counters shall negate any voltage spikes which occur when the voltage is removed from the counter.

3.3.4.2 Indicator lights. Indicator lights with individual dimming adjustments and press-to-test switch shall illuminate when 24-30 vdc is applied between pin R and pins C, X, and Y of connector A5J1.

3.3.4.3 Gunner-in-control (DS1). The blue light shall illuminate when 24-30 vdc is applied between pin D (positive) and pin R of connector A5J1.

3.3.4.4 40mm right, (DS2). The blue light shall illuminate when 24-30 vdc is applied between pin F (positive) and pin R of connector A5J1.

3.3.4.5 Armed (power) (DS3). The amber light shall illuminate when 24-30 vdc is applied between pin W (positive) and pin R of connector A5J1 with the weapon switch (S6) in the CLEAR position. The amber light shall illuminate also when 28 vdc is applied between pin G (positive) and pin R of connector A5J1 when (S6) is in the UNCLEAR position.

3.3.4.6 Turret power (DS4). The green light shall illuminate when 24-30 vdc is applied between pin J (positive) and pin R of connector A5J1.

3.3.4.7 7.62mm left (DS5). The blue light shall illuminate when 24-30 vdc is applied between pin $\overline{F}$ (positive) and pin R of connector A5J1.

3.3.4.8 40mm left (DS6). The blue light shall illuminate when 24-30 vdc is applied between pin E (positive) and pin R of connector A5J1.

3.3.4.9 Weapon unclear (DS10). The amber light shall illuminate when 24-30 vdc is applied between pin $\overline{A}$ (positive) and pin R of connector A5J1.

3.3.4.10 7.62mm right (DS11). The blue light shall illuminate when 24-30 vdc is applied between pin $\overline{B}$ (positive) and pin R of connector A5J1.

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3.4 Marking. Controls shall be marked in accordance with MIL-W-13855.

3.5 Workmanship. Workmanship shall be in accordance with MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. The first article (initial production unit(s)) shall be submitted for inspection in accordance with contract requirements (see 6.2). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein and such other inspection as is necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

4.5.1 Inspection lots. The formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105. Inspection lots shall be as large as practicable, in consideration of quality history, manufacturing conditions, the contractor's delivery schedule, and within the limitations of MIL-W-13855.

4.5.2 Examination.

4.5.2.1 Component parts and concurrent repair parts. Examination of parts shall be performed in accordance with the criteria specified in the contract (see 6.2). The contractor's examination of parts shall be accomplished prior to their assembly into the end items or submission for acceptance as repair parts.

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4.5.2.2 Controls. Each control shall be visually and manually examined as specified below. Controls failing to meet these requirements shall be rejected.

a. There are no missing, damaged or incorrectly assembled component parts.
b. There are no illegible, incorrect or missing product markings and identification plates (see 3.4).
c. Workmanship is in accordance with 3.5.

4.5.2.2.1 Weight. Examine the first three controls for compliance with the weight requirements specified in 3.2.1.

4.5.2.2.2 Final examination. Final examination of each control shall be performed after completion of testing and just prior to preservation and packaging. Controls shall be re-examined as specified in 4.5.2.2. Controls failing to meet these requirements shall be rejected.

4.5.2.3 Inspection approval stamp. Controls which have passed all examinations after successful completion of testing shall be stamped with the Department of Defense complete inspection approval stamp as specified on the applicable drawing. The Government representative will control the stamps used for such stampings to assure successful completion of all required examinations and tests.

4.5.3 Testing.

4.5.3.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete control. If test requirements cited herein are not met, acceptance of controls shall be deferred and the contractor shall accomplish as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional physical and visual examination of the components which are suspected to be the cause of failure or malfunction.
b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.
c. Examine controls, partially assembled controls, and components (including components and subassembles at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned control for retest.

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4.5.3.2 Functional testing. Each control shall be function tested (see 3.3) using the test method specified in 4.6.1. Controls failing to meet functional requirements shall be rejected.

4.5.3.3 Component parts and concurrent repair parts testing. Raw material testing, part testing, and certification shall be performed in accordance with the criteria specified in the contract (see 6.2). This will include chemical analysis and physical tests of materials, tests of protective finish, heat treatment, bonding, and function of parts as applicable. The contractor shall accomplish these tests prior to assembly of parts into the end item.

4.5.4 Packaging examination and testing. Unless otherwise specified (see 6.2), the packaging examination and testing shall be in accordance with MIL-P-14232.

4.5.5 Inspection equipment.

4.5.5.1 Acquisition, maintenance and disposition. Unless otherwise specified, responsibility for acquisition, maintenance and disposition of acceptance inspection and test equipment prescribed on the List of Inspection Equipment, Drawing IEL11687874 and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.5.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.6 Test methods.

4.6.1 Functional test. Controls shall be tested in accordance with the operating procedure specified in IEL11687874 (ref. Drawing 11836854).

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. The pilot pack shall consist of a complete control packaged in accordance with Packaging Data Sheet P11687874 for the level of protection specified in the contract (see 6.2), packed level C and forwarded in accordance with 3.1.

5.2 Preservation, packaging, packing and marking. Controls shall be preserved, packaged, packed and marked in accordance with Packaging Data Sheet P11687874 for the level of protection specified in the contract (see 6.2).

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5.3 Repair parts. Repair parts shall be preserved, packaged, packed and marked in accordance with the Packaging Data Sheet or the requirements, and for the level of protection specified in the contract (see 6.2).

6. NOTES

6.1 Intended use. The gunner control is intended for use on, but not limited to the M28A1 Helicopter Armament Subsystem.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. List of drawings and specifications pertinent to the gunner control showing applicable revision dates.
c. Requirement and number of controls to be submitted for first article by the contractor (see 3.1, 4.4 and 5.1).
d. Criteria for examination and testing of component parts and concurrent repair parts (see 4.5.2.1 and 4.5.3.3).
e. Packaging examination and testing, if different (see 4.5.4).
f. List of inspection equipment, responsibilities for acquisition, maintenance, and disposition of, if other than as specified (see 4.5.5.1).
g. Selection of applicable levels of preservation, packaging and packing (see 5.2).
h. Packaging instructions for repair parts (see 5.3).

6.2.2 Contract data requirements. When warrented, requirements for reports of examination and functional testing shall be as specified on a DD Form 1423 included in the contract.

6.2.3 Management control systems requirements. When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208 on the Management Control Systems Summary List, DD Form 1660.

6.2.4 Unless otherwise specified (see 6.2.1 f), the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

6.3 Supersession data. This specification includes the requirements of U.S. Army Weapons Command Purchase Description, USAWECOM PD-28, dated 13 March 1970 with Amendment 2, dated 31 March 1972.

Custodian:
Army - WC

Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-45924(WC); Control Panel, Gunner

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